

— DATA ANALYSIS REPORT

Public health data analysis in Cameroon

From raw data to diagnosis: cleaning, business indicators and actionable recommendations.

10,300 consultations · Year 2025

10 regions · Ministry of Health programme

Prepared by MAHOUSI DATA INSIGHTS · 19 July 2026

Executive summary

The essentials on one page

The analysis of **10,068 consultations** (after cleaning) reveals a broadly active system, yet undermined by a major supply problem: **18% of consultations face a medication stock-out**, with wide regional disparities (up to 20% in the East, Centre and South).

In addition, **nearly two in three patients (64.9%) are uninsured**, which weighs on access to care. The disease burden is dominated by **anemia** and **malnutrition**, while **chronic diseases** (hypertension, tuberculosis, diabetes) concentrate the highest treatment costs.

10,068

Consultations analysed

365,364

Total cost (USD)

18.0%

Medication stock-out rate

64.9%

Uninsured patients

Currency not specified in the source; expressed in USD by convention.

Data quality: what was fixed

The data received was "dirty", as is often the case in real conditions. Rigorous cleaning was carried out before any analysis, in R (tidyverse):

Issue detected	Treatment applied
232 strictly duplicated rows	Removed (10,300 → 10,068 rows)
Inconsistent regions (<i>SW, litoral, CENTER, Ctr...</i>)	Recoded to the 10 official regions
Inconsistent gender (<i>M, F, Masculin, male...</i>)	Harmonised to Male / Female
Aberrant ages (–5, 140, 222, and 537 zeros)	Replaced with missing value
Aberrant costs (–100, artificial ceiling at 50,000)	Replaced with missing value
Dates in 2 formats (YYYY-MM-DD and DD/MM/YYYY)	Unified into a single date
~5% missing values per column	Kept; each indicator computed on available data

Methodological choice: rather than deleting every row containing a missing value (which would have shrunk the sample), each indicator is computed on the observations actually available.

Analysis: medication stock-outs

Nearly **one consultation in five** faces a stock-out. The problem is national, but uneven across regions.

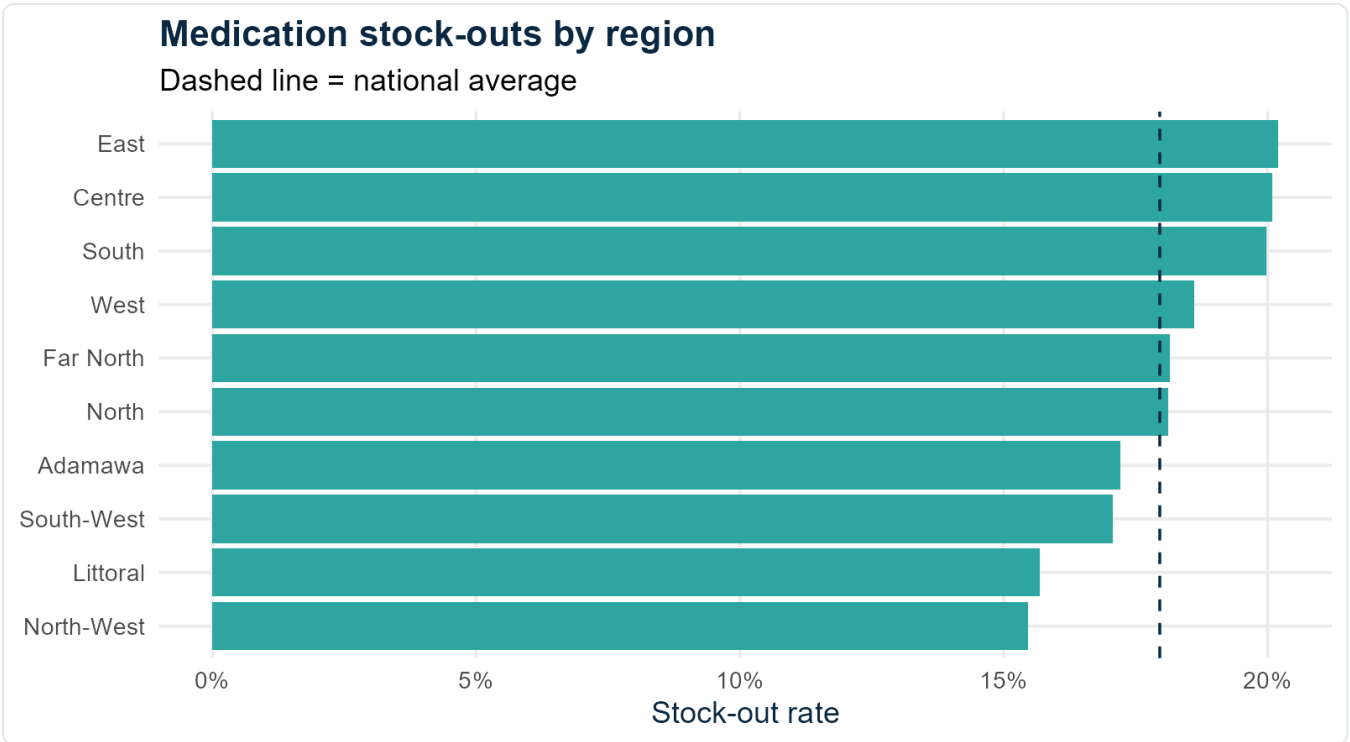


Figure 1 — Medication stock-out rate by region (dashed line = national average, 18.0%).

Region	Consultations	Stock-out rate	Level
East	955	20.2%	
Centre	956	20.1%	
South	961	20.0%	
West	946	18.6%	
Far North	981	18.1%	
North	922	18.1%	
Adamawa	1,023	17.2%	
South-West	920	17.1%	
Littoral	963	15.7%	
North-West	938	15.5%	

Analysis: disease burden

The burden is spread across 9 dominant conditions, led by **anemia** and **malnutrition** — two typical markers of nutritional vulnerability.

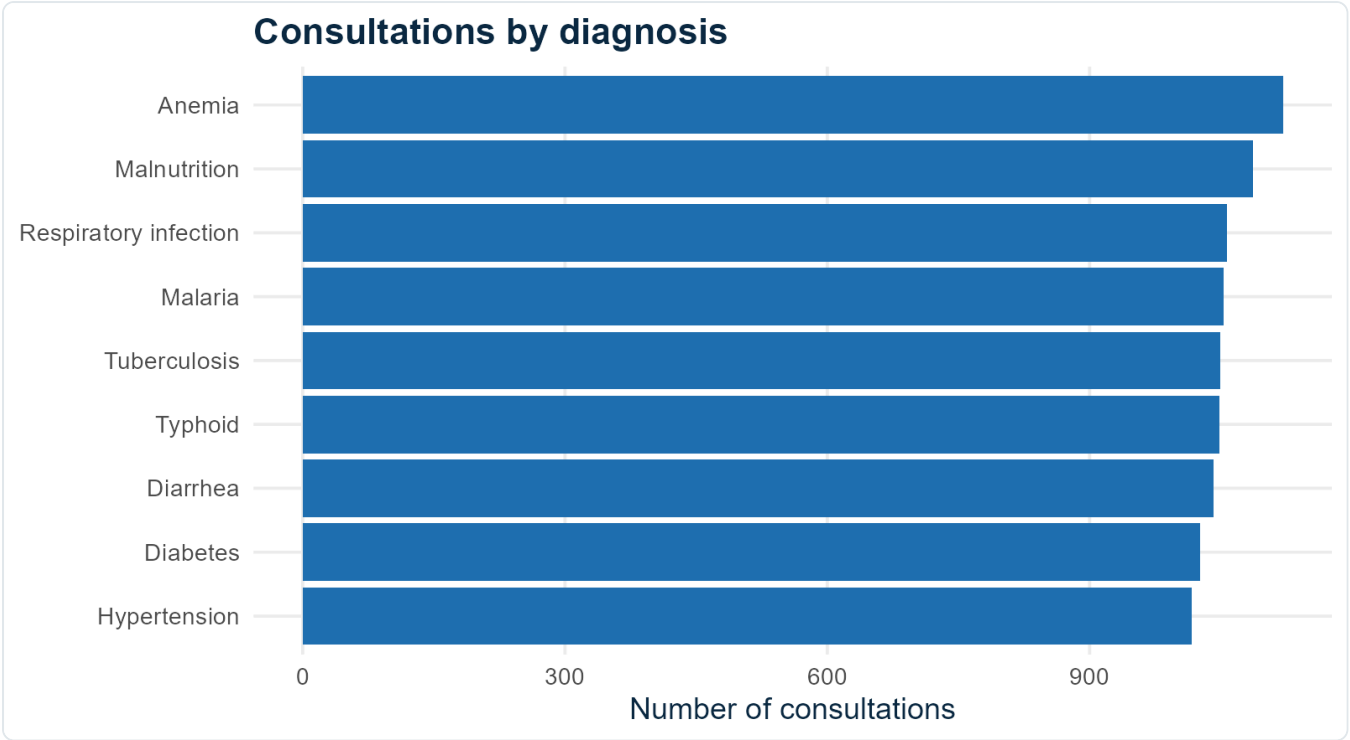


Figure 2 — Number of consultations by diagnosis.

Analysis: cost of care by condition

Chronic diseases (hypertension, tuberculosis, diabetes) are the most expensive to treat per patient — a long-term financing challenge.

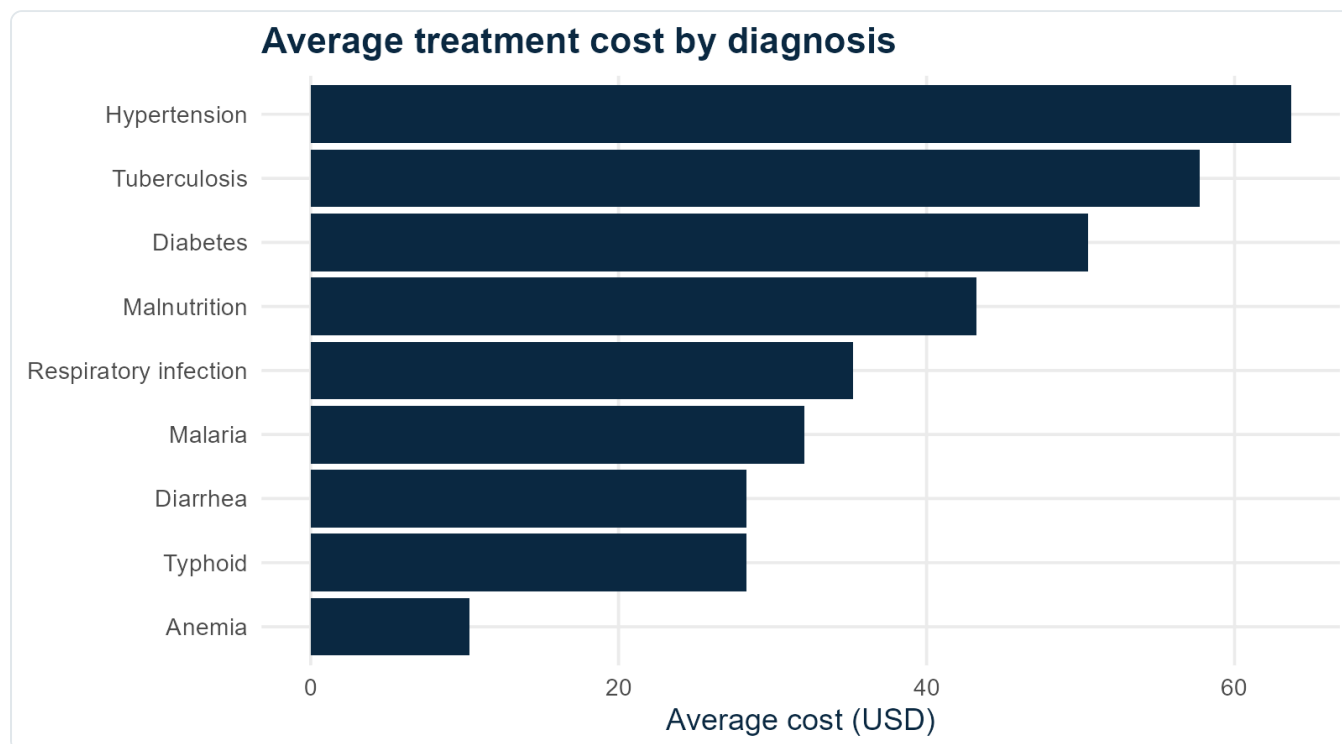


Figure 3 — Average treatment cost by diagnosis (USD).

Diagnosis	Mean cost	Median cost	Consultations
Hypertension	63.7	22.1	961
Tuberculosis	57.8	27.7	1,005
Diabetes	50.5	29.7	960
Malnutrition	43.2	14.3	1,030
Respiratory infection	35.3	14.8	990
Malaria	32.1	12.3	1,001
Diarrhea	28.3	7.8	988
Typhoid	28.3	18.2	992
Anemia	10.3	10.1	1,071

Analysis: insurance coverage

64.9% of patients are uninsured. High out-of-pocket costs can delay or prevent access to care, especially for the most expensive conditions.

Notably, **insured** patients show a higher average cost (45.5 vs 36.2 USD), a sign that they access more expensive care.

Status	Patients	Share	Mean cost
Uninsured	6,203	64.9%	36.2
Insured	3,362	35.1%	45.5

Analysis: trend and demographics

Activity is **stable** throughout the year (~800–880 consultations/month), with no marked seasonality in this sample. The average patient age is **31.7 years** (median 31, from 1 to 95 years).

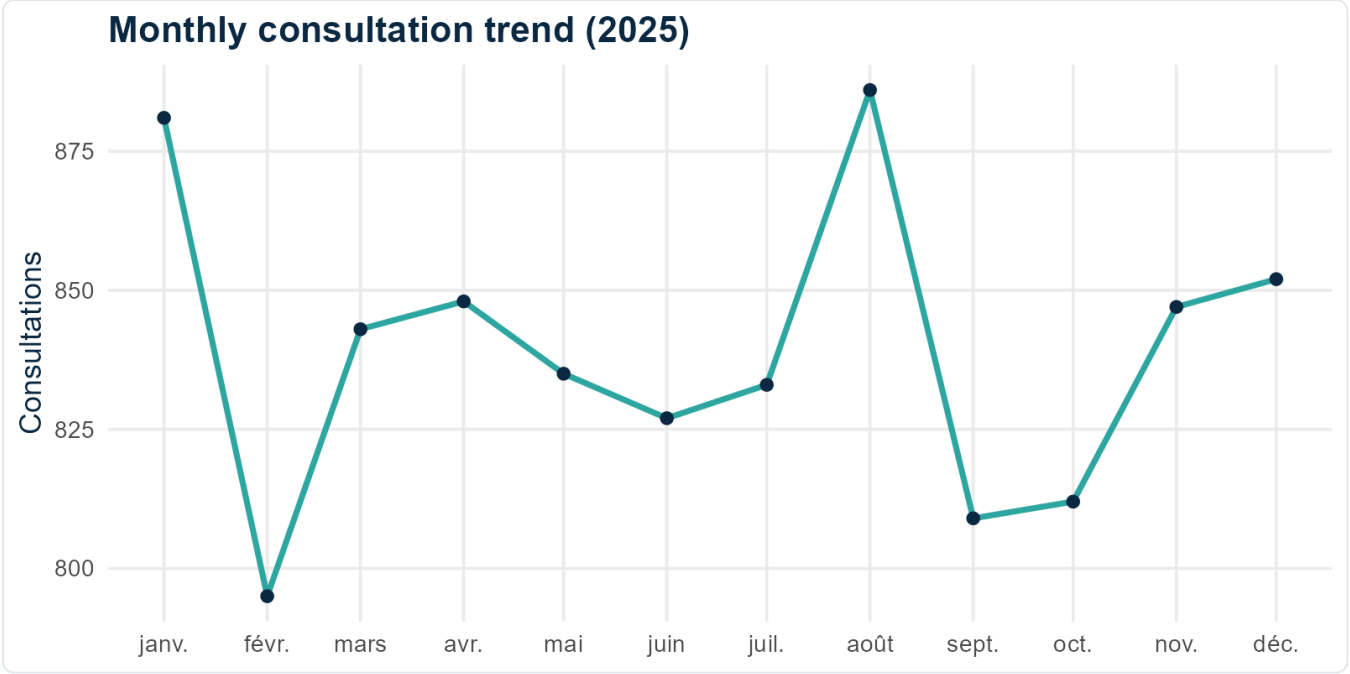


Figure 4 — Monthly consultation trend (2025).

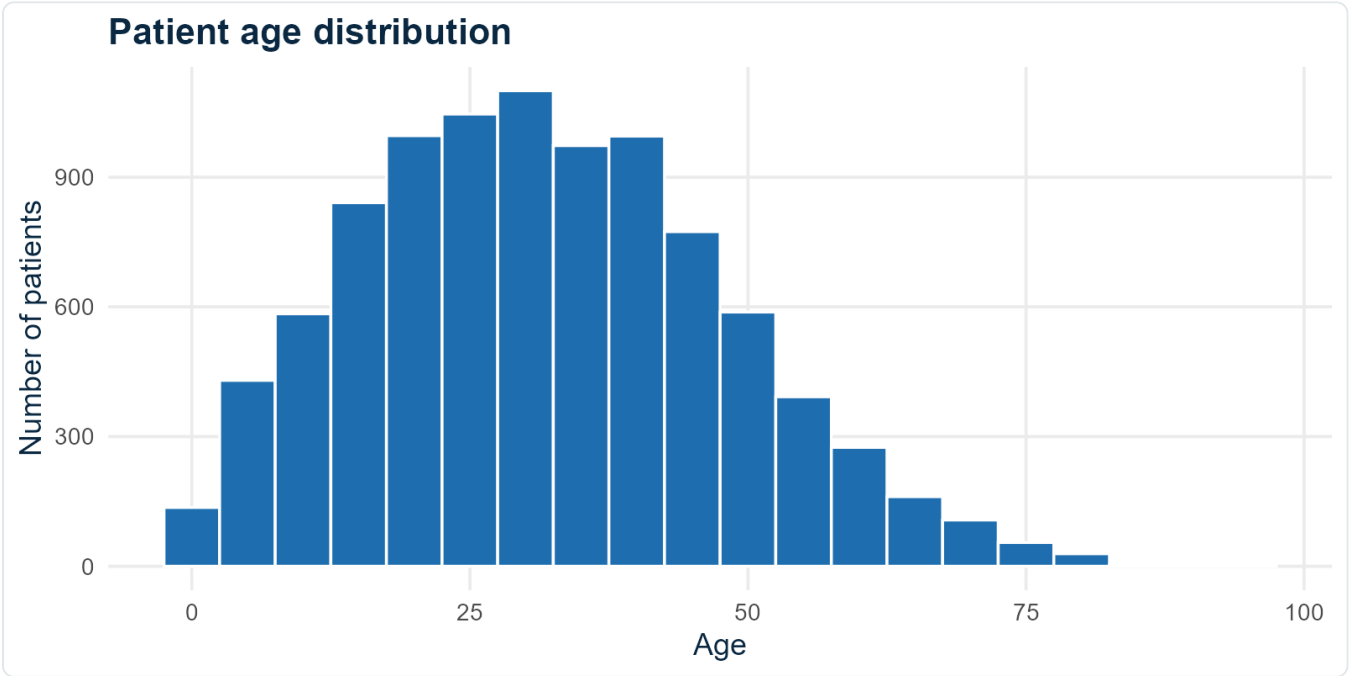


Figure 5 — Patient age distribution (after cleaning).

Recommendations: from data to decision

URGENT

1. Secure the medication supply chain

With 18% stock-outs, prioritise the East, Centre and South regions (>20%). Set up automatic replenishment thresholds, weekly stock monitoring, and a buffer stock in the most affected regions.

Owner: Pharmacy / logistics department · **Timeline:** 0–3 months

URGENT

2. Target anemia and malnutrition

Leading causes of consultation (≈23% combined). Launch nutrition campaigns, systematic screening and iron supplementation, prioritising the most affected age groups.

Owner: Nutrition programmes · **Timeline:** 0–6 months

STRUCTURAL

3. Anticipate the cost of chronic diseases

Hypertension, tuberculosis and diabetes weigh the most per patient. Plan long-term follow-up protocols and a dedicated budget for chronic treatments.

Owner: Financing / planning · **Timeline:** 6–12 months

STRUCTURAL

4. Expand insurance coverage

64.9% of patients are uninsured: out-of-pocket costs hinder access to care. Study a coverage scheme targeting first the most expensive conditions.

Owner: Ministry / health mutuals · **Timeline:** 6–18 months

STRUCTURAL

5. Improve data collection at the source

The errors (impossible ages, multiple date formats, mis-typed regions) come from data entry. Deploying dropdown lists and validity checks would avoid most future cleaning.

Owner: Health information system · **Timeline:** 3–9 months

Appendix — Methodology & reproducibility

This entire analysis was carried out in  (statistical programming language), from the raw file *dirty_health_data_subsaharan.csv* (10,300 rows). The pipeline is **fully reproducible**: re-running the script regenerates the clean file, the charts and this report identically.

1 — Import & cleaning (excerpt from *analyse_sante_cameroun.R*)

```
# Import and removal of exact duplicates
brut <- read_csv("dirty_health_data_subsaharan.csv", locale = locale(encoding = "UTF-8"))
donnees <- brut %>% distinct()

# Harmonise regions (SW, littoral, CENTER... -> 10 official regions)
donnees <- donnees %>% mutate(region_clean = case_when(
  str_to_lower(region) %in% c("littoral", "litoral") ~ "Littoral",
  str_to_lower(region) %in% c("sw", "sud ouest", "sud-ouest") ~ "Sud-Ouest",
  str_to_lower(region) %in% c("centre", "ctr", "center") ~ "Centre",
  region %in% regions_officielles ~ region,
  TRUE ~ NA_character_))

# Fix aberrant values (age and cost)
donnees <- donnees %>% mutate(
  age_clean = if_else(patient_age < 1 | patient_age > 120, NA_real_, patient_age),
  cost_clean = if_else(treatment_cost < 0 | treatment_cost >= 50000, NA_real_, treatment_cost))
```

2 — Computing the business indicators

```
# Medication stock-out rate by region
donnees %>% filter(!is.na(region_clean)) %>% group_by(region_clean) %>%
  summarise(taux_rupture = mean(medication_available == "Stockout", na.rm = TRUE)) %>%
  arrange(desc(taux_rupture))

# Average cost per diagnosis
donnees %>% filter(!is.na(diagnosis), !is.na(cost_clean)) %>% group_by(diagnosis) %>%
  summarise(cout_moyen = mean(cost_clean)) %>% arrange(desc(cout_moyen))
```

3 — Proof of execution: *sessionInfo()* output

```
R version 4.6.1 (2026-06-24 ucrt) – Platform: x86_64-w64-mingw32/x64

Attached packages:
tidyverse 2.0.0   ggplot2 4.0.3   dplyr 1.2.1   tidyr 1.3.2   readr 2.2.0
purrr 1.2.2   tibble 3.3.1   stringr 1.6.0   forcats 1.0.1
lubridate 1.9.5   janitor 2.2.1   scales 1.4.0
```

This *sessionInfo()* output is generated by R and certifies the versions actually used. Charts produced with **ggplot2**; interactive dashboard with **Shiny**; report laid out with **pagedown**.

Report produced with R (tidyverse) and pagedown · reproducible pipeline · © 2026 MAHOUSI DATA INSIGHTS — Turning Data into Better Decisions.